Will France Actually Win The EURO 2024?

MAT956: Advanced Mathematics for Data Science

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1 Introduction

The following report presents a data science project aimed at predicting the outcomes of the UEFA EURO 2024. By using historical match data and machine learining models, the project seeks to enhance the accuracy of football result predictions.

1.1 Motivation

Football generates great amounts of data that can be analyzed to predict match outcomes, contributing to a deeper understanding of the game. Predicting football results is extremely challenging due to the sport's unpredictability and numerous variables. In a simplified approach, we aim to tackle this challenge and get a model, that simulates the UEFA EURO 2024, predicts the winner of each match and eventually predicts the winner of the tournament.

1.2 Data

For the project the following two datasets were of essential use:

1.2.1 Match History

To predict the results of football matches, obviously the first necessary dataset required for learning was a one that held scorelines of past international matches of all the teams that participate in this years EURO's. The dataset we used contains every result of international football games from 1872 to 2024. For relevance reasons in this project only the matches from FIFA World Cups, UEFA European Chamionships and UEFA European Qualifiers were considered. This means international friendlies and tournaments in which no European nations participated were removed from the Dataset. International friendlies were removed due to them often being played at low stakes and thus do not give ideal results when trying to predict high pressure games in a European Championship The information left in the dataset was decided to be the following:

- index number of the match
- index number of the home team
- index number of the away team
- number of goals scored by home team
- number of goals scored by away team

- neutrality, meaning whether the match was played on neutral ground or not

1.2.2 Team Ratings

Next to charactirize each team by more than the name of the nation, we took the ratings from the most recent official FIFA video game (EAFC24). The amount of features for each team are extensive (it included even whether the team has a defensive/offensive playstyle, how well it performs from set pieces, in penalty shootouts, how well a team creates chances etc.) but we decided to keep just the core attributes:

- team name
- a unique index number
- overall, attack, midfield, and defence rating each a number between 1 and 100
- international prestige, a number between 1 and 10
- starting xi average age
- whole team average age

Unfrotunately there are nations participating at this years championship that have never appeared in any version of the FIFA videogame series, thus we had to manually add the following participants: Serbia, Georgia, Slovakia and Albania. Considering that they are missing from the videogames, due to them not always managing to qualify for the EURO's or the World Cup, we gave them ratings comparable to the lower end of European nations in EAFC2024, such as Bulgaria.

1.3 Merging the datasets

By indexing the teams in the historical matches dataset, we could plug in for each team the corresponding ratings obtained from the FIFA rating dataset. From this we obtained a dataset of 2492 football games where every match had 15 features (for the home and away team each: overall, attack, midfield and defense rating, international prestige, starting xi average age, whole team average age and additionally the neutrality parameter: 0 if the home team played with home advantage, 1 if the match took place on neutral ground).

2 Theory

Originally we planned to use a neural network, as football is highly tactical and depends on many different aspects. It was advised to us to try linear regression first, to have a benchmark for performance. However linear regression performed so much drastically better than a fully connected neural network, even when trying different layer sizes/numbers, activation functions, learning rates, that we decided to stick with linear regression. We briefly explain the theory behind it:

2.1 Linear Regression

Suppose the data $\mathcal{S} = (x_i, y_i)_{i=1}^n \subset \mathbb{R}^d \times \mathbb{R}^k$ is in a Euclidean space and let $\ell : \mathbb{R}^k \times \mathbb{R}^k \longrightarrow \mathbb{R}_{\geq 0}$ be a loss function. Define

$$\Theta_{d,k} = \{f : \mathbb{R}^d \longrightarrow \mathbb{R}^k, \text{s.t. } f(\boldsymbol{x}) = \boldsymbol{W}\boldsymbol{x} + \boldsymbol{c},\ \boldsymbol{c} \in \mathbb{R}^k, \boldsymbol{W} \in Mat_{k\times d}\}$$

to be the class of affine functions from from $\mathbb{R}^d$ to $\mathbb{R}^k$. Linear regression consists of finding $\hat{f}$ that minimizes the mean loss over the dataset:

$$\hat{f} := \underset{f\in\Theta_{d,k}}{\arg\min}\, \mathcal{L}(f) := \underset{f\in\Theta_{d,k}}{\arg\min}\, \frac{1}{n}\sum_{i=1}^{n} \ell(f(x_i), y_i)$$

In our case the loss function was chosen to be the squared loss $\ell(y, y') = ||y - y'||^2$, where $||x||$ is the standard Euclidean norm. This case of linear regression is particularly nice as it has a closed form solution, meaning there is no need to use optimization algorithms such as gradient descent.

3 Methodology

3.1 Training and testing

For training we would take $\frac{4}{5}$ of the merged set of historical matches and leave the remaining fifth for performance testing. The model we used was as already discussed linear regression with mean squared loss. The obtained predictor takes an array of 15 real values (corresponding to the ratings of home and away team and the neutrality parameter) and predicts an array of two values, namely the estimated game score (home and away goals).

3.2 Goal rounding

Since the linear regression does not predict exclusively integer values we had to decide for a way how to round the home and away goals. We felt however, purely rounding the result may not be ideal. Let us look at an example to illustrate our thought process. For the opening match between Germany and Scotland the linear regression model predicts that Germany would score an estimated 2.17 goals and Scotland 0.66. If we would round the results this would lead to a score of 2 : 1 in favor of Germany. However if we were to look at the estimated goal difference it would be 1.51, meaning the goal difference would be rounded to 2. But this would mean that Germany would win 2 : 0 or 3 : 1. So what would be the better prediction? That is hard to say, as football is very unpredictable; the team that is stronger on paper (Germany here) does not always win, it might underperform by only winning narrowly, drawing or even losing. To mimic the unpredictability we decided to do a fair coin toss:

- If it lands tails we procceed in the following way: First we check whether the estimated home or away goals number is smaller. We compute the estimated goal difference and then round the smaller of the two goal numbers. We also round the goal difference and add it to the rounded smaller goal number to obtain the larger goal number. In our example this means that since Scotlands score is smaller, it is rounded to 1 and then the rounded difference of 2 is added to 1 to obtain the goal number for Germany which is 3. This predicts the result 3 : 1
- If it lands heads instead we again check which one is the lower goal number. The lower one gets floored and the larger one gets rounded. The idea is that this way close matches that should end in a draw, still end up in a draw, but matches were there is a clear favourite should still get weighted.

4 Results

4.1 Performance results

The mean squared loss on the test set was approximately 1.55. This means that on average the norm of the preducted home and away score deviated from the correct result by about 1.5 goals. At first this might not look very accurate, but considering that precise score results in football matches are very difficult, being off by 1.5 total goals is not too bad.

4.2 EURO 2024 prediction results

Below one can see the results for the group stage game, aswell as the standings for the group stage including points and goal differences:

	home_number	away_number	home_team	away_team	home_score	away_score	neutral
2492	2	16	Germany	Scotland	2.0	0.0	0
2493	21	39	Hungary	Switzerland	1.0	1.0	1
2494	5	9	Spain	Croatia	2.0	1.0	1
2495	4	59	Italy	Albania	3.0	1.0	1
2496	49	10	Slovenia	Denmark	1.0	2.0	1
2497	56	0	Serbia	England	0.0	2.0	1
2498	12	7	Poland	Netherlands	0.0	2.0	1
2499	31	1	Austria	France	1.0	2.0	1
2500	25	19	Romania	Ukraine	1.0	1.0	1
2501	8	58	Belgium	Slovakia	2.0	1.0	1
2502	40	57	Turkey	Georgia	1.0	1.0	1
2503	3	14	Portugal	Czech Republic	2.0	0.0	1
2504	2	21	Germany	Hungary	2.0	0.0	0
2505	16	39	Scotland	Switzerland	1.0	1.0	1
2506	9	59	Croatia	Albania	2.0	0.0	1
2507	5	4	Spain	Italy	1.0	1.0	1
2508	49	56	Slovenia	Serbia	1.0	2.0	1
2509	10	0	Denmark	England	1.0	1.0	1
2510	12	31	Poland	Austria	1.0	1.0	1
2511	7	1	Netherlands	France	1.0	1.0	1
2512	58	19	Slovakia	Ukraine	1.0	2.0	1
2513	8	25	Belgium	Romania	2.0	1.0	1
2514	57	14	Georgia	Czech Republic	1.0	2.0	1
2515	40	3	Turkey	Portugal	1.0	2.0	1
2516	2	39	Germany	Switzerland	2.0	0.0	0
2517	16	21	Scotland	Hungary	1.0	1.0	1
2518	59	5	Albania	Spain	0.0	2.0	1
2519	9	4	Croatia	Italy	1.0	1.0	1
2520	0	49	England	Slovenia	2.0	1.0	1
2521	10	56	Denmark	Serbia	1.0	1.0	1
2522	7	31	Netherlands	Austria	2.0	1.0	1
2523	1	12	France	Poland	2.0	1.0	1
2524	58	25	Slovakia	Romania	1.0	1.0	1
2525	19	8	Ukraine	Belgium	1.0	1.0	1
2526	57	3	Georgia	Portugal	0.0	2.0	1
2527	14	40	Czech Republic	Turkey	1.0	1.0	1

Figure 1: Predicted match results for group stage

Points Table with Goal Difference:

	Group:	Points	GoalsScored	GoalsConceded	GoalDifference
Portugal	F	9	6.0	1.0	5.0
Czech Republic	F	4	3.0	4.0	-1.0
Turkey	F	2	3.0	4.0	-1.0
Georgia	F	1	2.0	5.0	-3.0
Belgium	E	7	5.0	3.0	2.0
Ukraine	E	5	4.0	3.0	1.0
Romania	E	2	3.0	4.0	-1.0
Slovakia	E	1	3.0	5.0	-2.0
Netherlands	D	7	5.0	2.0	3.0
France	D	7	5.0	3.0	2.0
Austria	D	1	3.0	5.0	-2.0
Poland	D	1	2.0	5.0	-3.0
England	C	7	5.0	2.0	3.0
Denmark	C	5	4.0	3.0	1.0
Serbia	C	4	3.0	4.0	-1.0
Slovenia	C	0	3.0	6.0	-3.0
Spain	B	7	5.0	2.0	3.0
Italy	B	5	5.0	3.0	2.0
Croatia	B	4	4.0	3.0	1.0
Albania	B	0	1.0	7.0	-6.0
Germany	A	9	6.0	0.0	6.0
Scotland	A	2	2.0	4.0	-2.0
Hungary	A	2	2.0	4.0	-2.0
Switzerland	A	2	2.0	4.0	-2.0

Figure 2: Group stage final standings prediction

It is important to mention that our predictions were made before the start of the EURO's and were never updated after new results. From this we extract the first and second placed teams from each group, aswell es the four best performing third place teams:

1. Teams that qualified as first seed: Germany, Spain, England, Netherlands, Belgium, Portugal
2. Teams that qualified as second seed: Switzerland, Italy, Denmark, France, Ukraine, Czech republic
3. Teams that qualified as best third seeds (2 points and -1 goal difference or better): Croatia, Serbia, Romania, Turkey.

Switzerland was selected as second seed because despite havig equal results when compared to Scotland and Hungary, because out of the three they had the highest team rating. The playoff bracket was then

constructed according to the UEFA guidelines corresponding to our scenario. The predicted results for the first round of the knock-out stage are as follows:

	home_team	away_team	home_score	away_score	neutral
0	Spain	Turkey	2.0	0.0	1
1	Germany	Denmark	2.0	0.0	0
2	Portugal	Croatia	2.0	1.0	1
3	France	Ukraine	2.0	1.0	1
4	Belgium	Serbia	1.0	1.0	1
5	Netherlands	Czech Republic	2.0	1.0	1
6	England	Romania	2.0	0.0	1
7	Switzerland	Italy	0.0	2.0	1

Figure 3: Round of 16 predictions

The match between Belgium and Serbia ended in a draw, thus we coded a penalty shootout decided function. It is essentially a Bernoulli trial were the probabilty for a team to win it is proportional to their estimated goals from the linear regression. The Bernoulli trial let Belgium win, which seems reasonable as they had a slightly larger goal expectency (approximately 1.67 compared to 1.26). We continue with the results for the quarter finals:

	home_team	away_team	home_score	away_score	neutral
0	Spain	Germany	1.0	2.0	1
1	Portugal	France	1.0	1.0	1
2	Belgium	Netherlands	1.0	1.0	1
3	England	Italy	1.0	1.0	1

Figure 4: Quarter finals prediction

This time there are three draws, and the penalties went they way of Portugal, Netherlands and England. Next for the seminfinals we have the following prediction:

	home_team	away_team	home_score	away_score	neutral
0	Germany	Portugal	2.0	1.0	0
1	Netherlands	England	1.0	1.0	1

Figure 5: Semifinals predition

The penalty shootout went in favor of England. Thus the final is predicted to be between England and Germany. The final prediction is:

```
print(f"The score of the final Germany - England is {int(round_final[0])}:{int(round_final[1])}! ")
print("The winner of the UEFA Euro 2024 is Germany!")
```

```
The score of the final Germany - England is 2:1!
The winner of the UEFA Euro 2024 is Germany!
```

Figure 6: Finals prediction

5 Discussion

5.1 Model issues

While trying the predictions on different imaginary teams, we tried letting very good teams (with ratings higher than 90) play against very bad teams (ratings lower than 50). The model predicted a negative score for the lower rated team. The issue lies therein that the prediction model is linear and thus the possible score lines are not bounded from below. During the EURO simulations this does not interfere with the results since teams rated as low than that did not qualify for the tournament. Therefore negative results did not come up in the simulations and due to that, we decided, not to adjust the code.

Another issue when using the linear regression model, is that certain parameters contribute in a monotonic way to the goal results. As for example one will see in the parameter testing section 5.3 that the model seems to think that the higher the squads average age, the better. This is not true at all in real life (obviously), as for a football team neither a too low average age is good (leads to lack of experience) nor a too old average age (leads to lack of fitness/more injury prone).

Similarly the neutrality parameter adds a flat ≈ 0.54 expected goals to the home team, when it is 0 (meaning when the home team plays in their own country) compared to when the neutrality parameter is 1. This is not reasonable either as it should not add the same expected goal amount to a team that is rather weak (for example Liechtenstein) compared to a strong team (for example Germany).

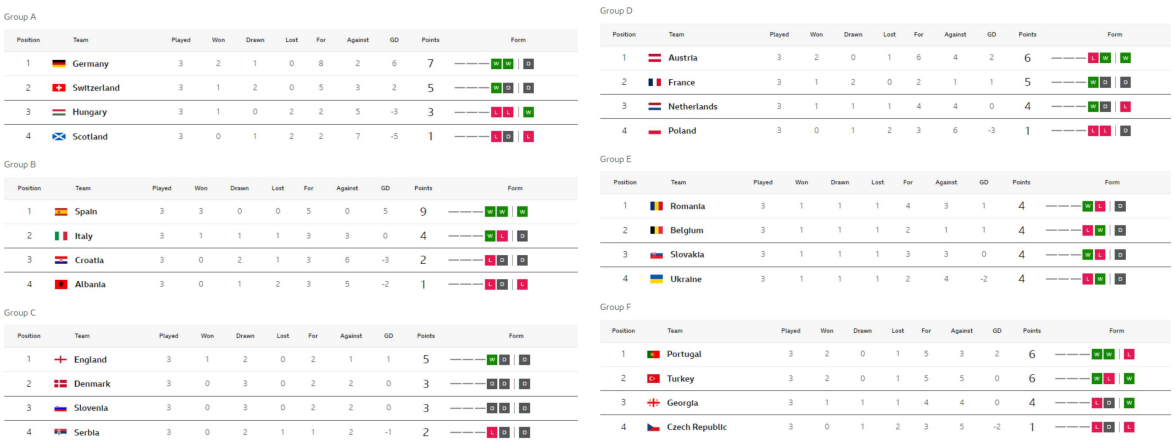
5.2 Predicted vs. real outcome of the group stage

How reasonable were the predictions of the group stage matches? If one would ask the average football fan, the linear model predicts a very reasonable result. All of the historically strong nations (Germany,

France, Spain, Netherlands, England, Italy, Portugal etc) are expected to make it out, in most cases as first seed. It also expects nations to perform well that only in recent history surged as strong teams such as Croatia, Belgium, Denmark. The teams that are predicted to end last in their group are historically weaker nations such as Slovakia, Albania, Georgia.

It is far from perfect however, for example Austria is predicted to get only one point despite being in good form leading up to the tournament since appointing experienced coach Ralf Rangnick. Switzerland is also expected to only achieve only 2 points despite not missing a single EUROs or World Cup knockout stage since 2010. It also does not predict a match with a goal difference higher than 2, which is hardly realistic. The issue could be that historically even though matches with very one sided results such as 3 : 0, 4 : 1 or even 4 : 0 do arise multiple times in a tournament, they are far less frequent than matches with only 2 goals overall. Hence our models much more favours a lot results with less goals and less one sided score lines.

Let us now look at the real standings for the group stage after all the games took place:



Group A

Position	Team	Played	Won	Drawn	Lost	For	Against	GD	Points	Form
1	Germany	3	2	1	0	8	2	6	7	W W D
2	Switzerland	3	1	2	0	5	3	2	5	W D D
3	Hungary	3	1	0	2	2	5	-3	3	L L W
4	Scotland	3	0	1	2	2	7	-5	1	L D L

Group B

Position	Team	Played	Won	Drawn	Lost	For	Against	GD	Points	Form
1	Spain	3	3	0	0	5	0	5	9	W W W
2	Italy	3	1	1	1	3	3	0	4	W L D
3	Croatia	3	0	2	1	3	6	-3	2	L D D
4	Albania	3	0	1	2	3	5	-2	1	L D L

Group C

Position	Team	Played	Won	Drawn	Lost	For	Against	GD	Points	Form
1	England	3	1	2	0	2	1	1	5	W D D
2	Denmark	3	0	3	0	2	2	0	3	D D D
3	Slovenia	3	0	3	0	2	2	0	3	D D D
4	Serbia	3	0	2	1	1	2	-1	2	L D D

(a) Groups A-C

Group D

Position	Team	Played	Won	Drawn	Lost	For	Against	GD	Points	Form
1	Austria	3	2	0	1	6	4	2	6	L W W
2	France	3	1	2	0	2	1	1	5	W D D
3	Netherlands	3	1	1	1	4	4	0	4	W D L
4	Poland	3	0	1	2	3	6	-3	1	L L D

Group E

Position	Team	Played	Won	Drawn	Lost	For	Against	GD	Points	Form
1	Romania	3	1	1	1	4	3	1	4	W L D
2	Belgium	3	1	1	1	2	1	1	4	L W D
3	Slovakia	3	1	1	1	3	3	0	4	W L D
4	Ukraine	3	1	1	1	2	4	-2	4	L W D

Group F

Position	Team	Played	Won	Drawn	Lost	For	Against	GD	Points	Form
1	Portugal	3	2	0	1	5	3	2	6	W W L
2	Turkey	3	2	0	1	5	5	0	6	W L W
3	Georgia	3	1	1	1	4	4	0	4	L D W
4	Czech Republic	3	0	1	2	3	5	-2	1	L D L

(b) Groups D-F

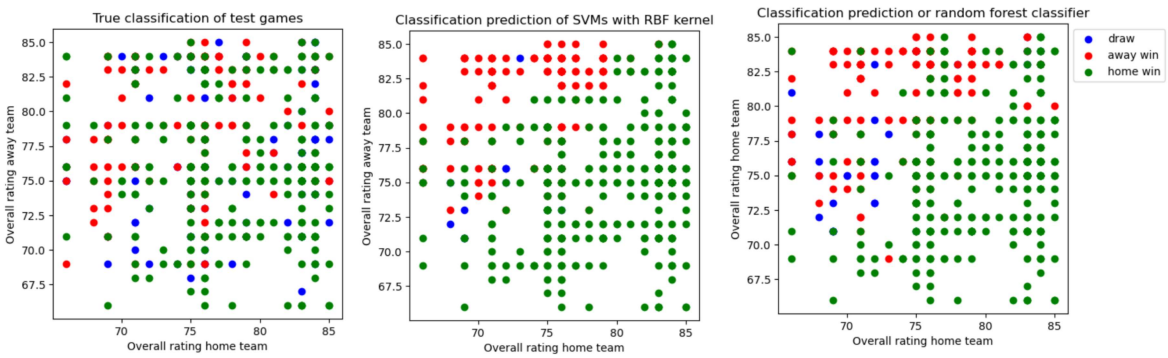
Figure 7: Final group stage standings, credit: BBC official website

We can see that the real results do not differ too much from our prediction. For example in 4 out of 6 groups the first placed teams coincide, with the exceptions being Austria and Romania placing first in Group D and Group E respectively, both being unexpected results as many would have thought Netherlands or France and Belgium taking the top spot in their respective group (despite Austria being in good form). Worth to mention is that the rankings in Group B were perfectly predicted, only

differing by points and goal difference. Overall out of the 16 teams that our regression predicted to make it out of the group stage, 13 actually made it through but in some cases differing from which position (first, second or third). When looking at the matches directly 19 out of 36 games were predicted correctly between home victory, draw and away victory, meaning an accuracy of approximately 52.8%. While this does seem low, it still performs a lot better than a random guesser (which we would expect to have 33.3% accuracy).

When it comes to correct score lines we see that our model is rather weak, only in 10 out of 36 group stage games had the correct goal difference, in even less matches did the score precisely coincide. As already mentioned it could not predict a score line as high as Germany Scotland 5 : 1 or Romania beating Ukraine 3 : 0. It also could not predict upsets such as Slovakia beating Belgium and Austria beating the Netherlands.

All in all for a model as simple as linear regression with mean squared loss, in our view this is good performance, when keeping in mind the highly unpredictable nature of football and sports in general. To illustrate this unpredictability we also tried to view the the matches as classification problem (class 0 = match ends in a draw, class 1 = match ends in home win, class 2 = match ends in away win), and tried to classify the games using SVMs (Support Vector Machines) and Random Forest algorithms. We used the same merged dataset as before with a 4 : 1 split and looked at the performance in the test set. Over several trials the accuracy ranged between 49.5% and 56.4% for both classifiers, so pretty much the same as the accuracy obtained from our linear regression model. The unpredictability of football games can be seen very nicely in our data. Below one can see the games in the test set when only looking at the overall team rating (which takes in consideration attack, midfield and defense rating) of the home team vs the overal rating of the away team. The first image shows the true labels, the second shows the prediction of the SVM with RBF Kernel and the third image shows the prediction of the Random Forest Classifier:



We can see in the first image which has the true label that the data is very noisy, meaning there are many "outliers" that represent upsets where a weaker team draws or even wins against a much stronger team. For example in the bottom right area of the first image, where the away team has far lower rating compared to the home team, that there is still several draws or away wins happening (represented red and blue dots). This is why both classifiers struggle with finding a good decision boundary as one can see in the second and third image, because there is always going to be upset games, while the classifiers as expected but the decision boundaries at the around the diagonals predicting wins for away or home team depending which rating is higher and occasionally a draw when the ratings are similar.
A small part of the noise could come from the merging of our datasets, as for matches that took place decades the ratings do not coincide with the corresponding team. For example Croatia is much better today than it was many years ago. However the international prestige parameter takes this into consideration, and most of the nations that were historically good (or bad) still are among the best in this tournament.

5.3 Testing impact of feature parameters

Since several factors were taken into account in the model, it was of interest, which factors had an impact on the results. For this we looked at some of the factors seperately. For this we always would vary either the international prestige, the four rating parameters or the age, for the away team while keeping the all the other parameters fixed and equal between home and away team. We did this each time both for the case when the match is played on neutral ground and for the case when not. The specific slope for each of the curve can be seen in figure

5.3.1 Impact of International Prestige

We looked at the match outcome of two teams while iterating the international prestige for the home team from 1 to 10. As can be seen in the figure below, the higher the prestige is, the bigger the expected goal difference gets (as we would have hoped).

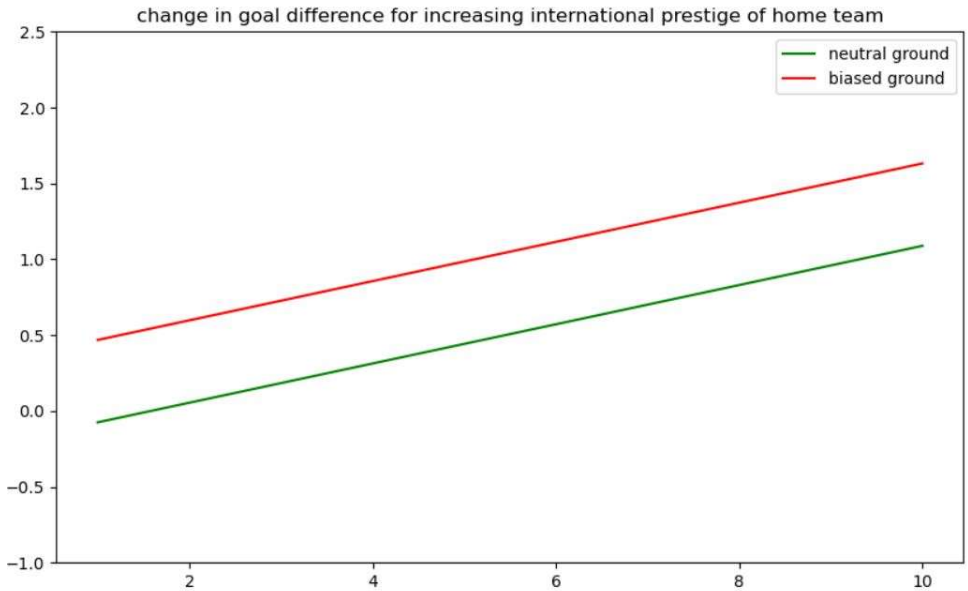


Figure 9: x-axis: International prestige, y-axis: expected goal difference

5.3.2 Impact of team rating

Here we would vary all the ratings at the same time from 50 to 100 for the home team. As expected, the higher the rating the more favourable is the goal difference for the home team.

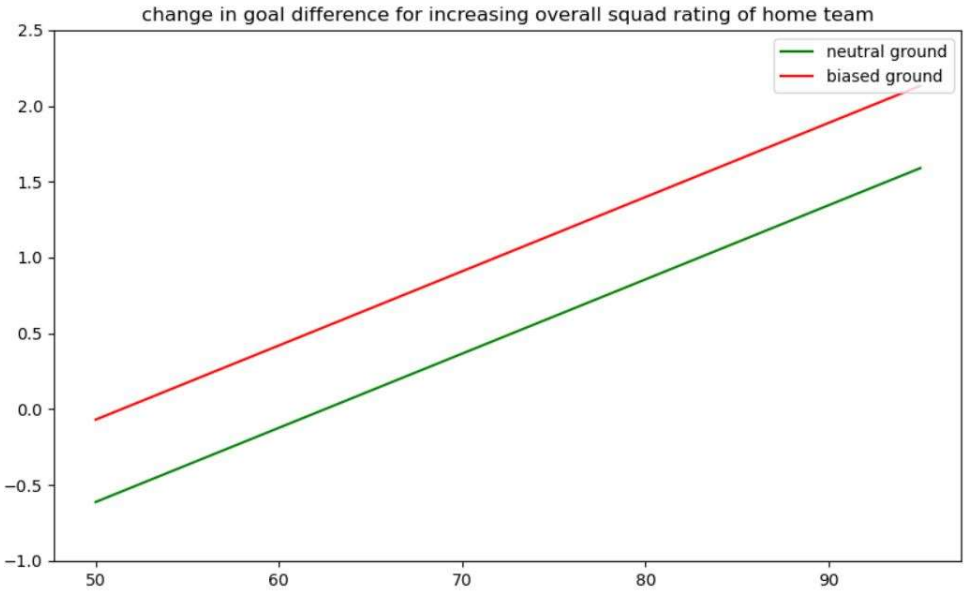


Figure 10: x-axis: squad rating (overall, att, midf, def), y-axis: expected goal difference

5.3.3 Impact of average age

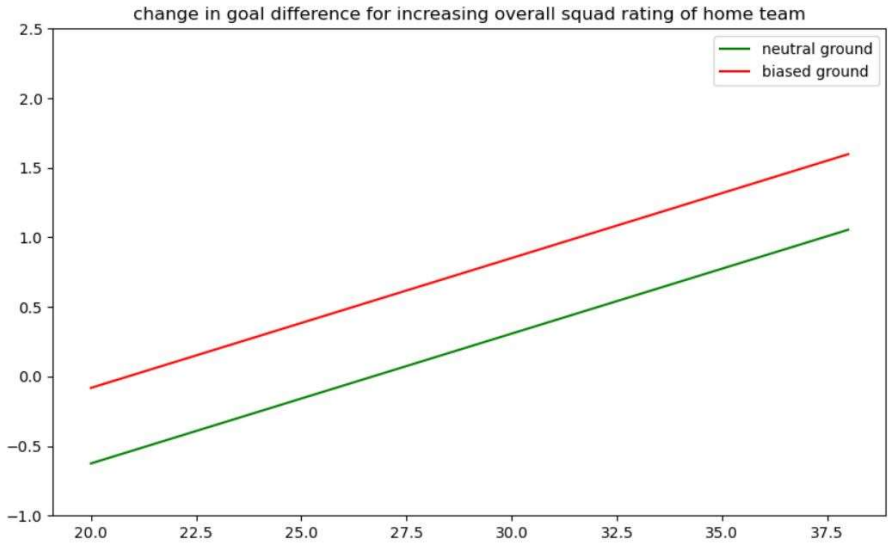


Figure 11: x-axis: average age of overall squad, y-axis: expected goal difference

Finally we also varied the average age of the home team from 20 to 38, which can be seen in the figure above. Here we see a limit to our linear model, as it can only try to fit a linear relationship between age and expected goal difference, meaning either the lower the average age the worse, or the better, which as already explained does not reflect real life. We also see that the expected goal difference in each figure differs between neutral and biased ground always by a fixed value of approximately 0.54 in favor of the home team.

Finally we may also look at the slope of each of the curves in the three images, to get a grasp to which parameters the model is more sensitive to.

```
slope_p = (test_diff_p_1[-1] - test_diff_p_1[0])/9
print(f'the slope of the international prestige parameter is: {slope_p}')
slope_r = (test_diff_r_1[-1] - test_diff_r_1[0])/50
print(f'the slope of the overall rating parameter is: {slope_r}')
slope_a = (test_diff_a_1[-1] - test_diff_a_1[0])/18
print(f'the slope of the age parameters is: {slope_a}')
```

```
the slope of the international prestige parameter is: 0.1292762148037577
the slope of the overall rating parameter is: 0.044050425625635464
the slope of the age parameters is: 0.09333258749728096
```

Figure 12: Slopes of the different parameter curves

6 Next Steps and Conclusion

To further refine the model, the problem of negative scores explained in Section 5 should be adjusted. A quick way of doing this is shifting the scores to get positive values and keeping the goal difference the same. For example if the model predicts an outcome of $2 : -1$ the adjusted score would then be $3 : 0$. Like this the model would not fail if teams with a big difference in their ratings play against each other.

As previously noted, match outcomes are influenced by many variables beyond team rankings and match history alone. Enhancing the introduced model could involve refining it with additional factors, such as analyzing the distance each team runs during a match or considering the number of shots on target. Another potential enhancement could involve evaluating players from top leagues who score the most goals during matches, and incorporating their expertise accordingly.

As the model takes more factors into account, its accuracy is likely to improve. The successful application of machine learning to predict football match outcomes could revolutionize sports analytics. However, it's important to understand that even the most advanced program will never achieve perfect prediction of every match outcome.

And finally to answer the question stated in the title: No, they won't. Which is, after watching the tournament so far, not surprising.

7 References

1. EAFC 2024 team ratings dataset provided by Stefano Leone on kaggle:
   https://www.kaggle.com/datasets/stefanoleone992/ea-sports-fc-24-complete-player-dataset
2. International football results dataset from 1872 to 2024 provided by Mart Jürisoo on kaggle:
   https://www.kaggle.com/datasets/martj42/international-football-results-from-1872-to-2017
3. EURO 2024 group stage results, taken from official BBC website:
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